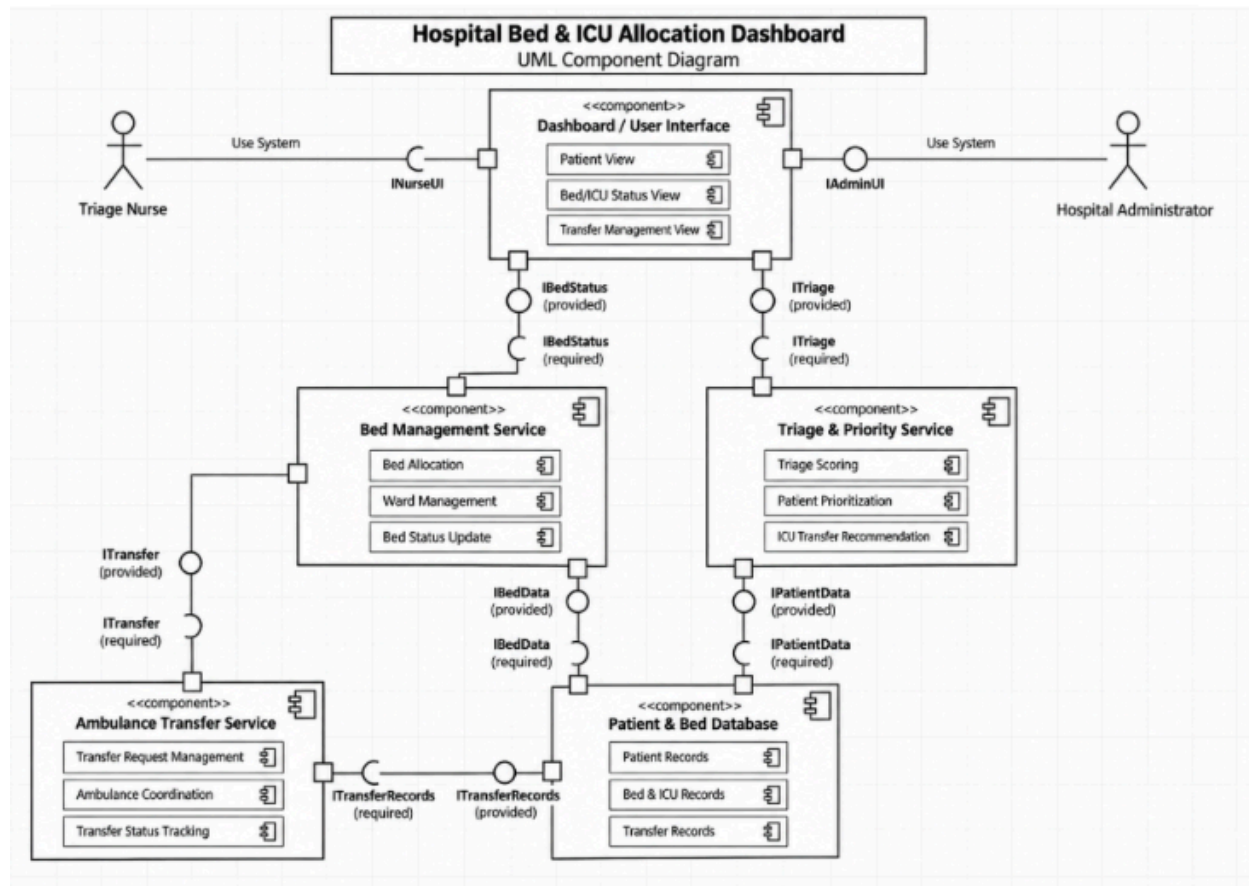


Hospital_Bed_ICU_Component_Diagram



Architecture Selection and Justification

Hospital Bed & ICU Allocation Dashboard

Architecture Selection:

We chose **Microservices Architecture** for the Hospital Bed & ICU Allocation Dashboard.

Architectural Choice:

The system is divided into independently deployable services such as Bed Management, Triage & Priority, Ambulance Transfer, Dashboard, and Patient & Bed Database services.

Reason 1 – Independent Scaling:

The dashboard must handle real-time ward occupancy and ICU availability. With microservices,

heavily used services such as Bed Management and Triage & Priority can be scaled independently without scaling the entire system.

Reason 2 – Fault Isolation:

The system supports critical healthcare operations. If the Ambulance Transfer Service temporarily fails, other services such as bed monitoring and triage prioritization can continue operating.

Security Advantage:

Microservices allow security controls to be applied separately to different services. Sensitive patient information can be protected through authentication, authorization, and controlled access between services.

Performance Benefit:

The Bed Management Service can provide real-time bed availability independently, allowing the dashboard to receive updated occupancy information quickly. Independent scaling also helps maintain performance during peak emergency-room loads.